DATABASE SPECIFICATION / DATA DICTIONARY

MEDSYS Database Dictionary

Detailed database reference for the MEDSYS backend, covering tables, columns, defaults, primary keys, foreign keys, uniqueness, nullability, indexes, check constraints, enums, and partitioning behavior.

DOCUMENT TYPE	SYSTEM
As-built relational database dictionary	MEDSYS API Enterprise Backend
VERSION	**DATE**
1.2	March 16, 2026
SOURCE OF TRUTH	**AUDIENCE**
Flyway migrations V1, V4, V5, V6, V7, V8, and V9	Developers, QA, BA, reviewers, and client stakeholders

Prepared from the current repository schema migrations. This document describes the implemented database design, not a proposed future-state model.

1. Scope and Reading Guide

This document provides a field-level dictionary for the MEDSYS PostgreSQL schema. It focuses on database-enforced validation and structure: data type, required versus nullable, default values, generated values, primary keys, unique constraints, foreign keys, indexes, and check constraints.

> **Important:** application-level validations may be stricter than the database. This document records what the database itself enforces. Migration `V8` adds client/demo seed data, while `V9` adds user-level extra permission storage for effective authorization overrides.

2. Global Design Rules

- Most business tables are organization-scoped through `organization_id`.
- Most tables use surrogate numeric primary keys via `BIGSERIAL` or a sequence-backed `BIGINT`.
- Many business tables use soft deletion through nullable `deleted_at`.
- Appointments and audit logs are partitioned for operational scale.
- Authentication uses refresh-token family tracking for replay detection.

3. Enum Types

ENUM TYPE	ALLOWED VALUES	USED BY
user_role	`owner`, `doctor`, `assistant`	`users.role`
gender	`male`, `female`, `other`	`patients.gender`
allergy_severity	`low`, `moderate`, `high`	`patient_allergies.severity`
appointment_status	`waiting`, `in_consultation`, `completed`, `cancelled`	`appointments.status`
priority_level	`low`, `normal`, `high`, `critical`	`appointments.priority`
test_order_status	`ordered`, `in_progress`, `completed`, `cancelled`	`test_orders.status`
drug_source	`clinical`, `outside`	`prescription_items.source`
inventory_category	`medicine`, `consumable`, `equipment`, `other`	`inventory_items.category`
inventory_movement_type	`in`, `out`, `adjustment`	`inventory_movements.movement_type`

4. Relationship Summary

PARENT TABLE	CHILD TABLE	RELATIONSHIP
users	`refresh_tokens`	One user to many refresh tokens via `refresh_tokens.user_id`.
families	`patients`, `family_members`	Families optionally attach directly to patients and also map through a join table.
patients	`family_members`, `patient_allergies`, `patient_conditions`, `appointments`, `encounters`, `prescriptions`, `patient_vitals`, `patient_timeline_events`, `patient_history_entries`	Patient is the central clinical entity and is referenced across most domain tables.
appointments	`encounters`	One encounter per appointment, enforced by unique `encounters.appointment_id` and a composite FK to partitioned appointments.

PARENT TABLE	CHILD TABLE	RELATIONSHIP
encounters	`encounter_diagnoses`, `test_orders`, `prescriptions`, `patient_vitals`, `patient_timeline_events`	Encounter is the anchor for diagnoses, test orders, prescriptions, and some clinical observations.
prescriptions	`prescription_items`, `dispense_records`	One prescription to many items and many dispense events.
inventory_items	`inventory_movements`	One inventory item to many immutable movement rows.
users	`appointments`, `encounters`, `prescriptions`, `dispense_records`, `inventory_movements`, `patient_history_entries`, `audit_logs`	Users appear as operational actors throughout the system.

5. Core Security and Identity Tables

5.1 users

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant/organization scope key.
`uuid`	UUID	Yes	`gen_random_uuid()`	Unique row UUID.
`email`	CITEXT	Yes	None	Case-insensitive email; unique with `organization_id`.
`password_hash`	TEXT	Yes	None	Hashed password only.
`first_name`	VARCHAR(80)	Yes	None	Required given name.
`last_name`	VARCHAR(80)	Yes	None	Required surname.
`role`	user_role	Yes	None	Enum constrained.
`extra_permissions`	JSONB	Yes	`'[]'::jsonb`	Stores per-user permission overrides; added in `V9`.
`is_active`	BOOLEAN	Yes	`TRUE`	Operational active flag.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.

Constraints: primary key on `id`; unique on `uuid`; unique on `(organization_id, email)`. **Notes:** `extra_permissions` is a required JSONB array column with an empty-array default after `V9`.

5.2 refresh_tokens

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`user_id`	BIGINT	Yes	None	Foreign key to `users.id`.
`token_id`	UUID	Yes	`gen_random_uuid()`	Unique token identifier.
`family_id`	UUID	Yes	None at DB level	Refresh-token family identifier; added in V6.
`parent_token_id`	UUID	No	None	Logical previous token pointer; no FK constraint.
`expires_at`	TIMESTAMPTZ	Yes	None	Expiration timestamp.
`used_at`	TIMESTAMPTZ	No	None	Marked when refresh token is consumed.
`revoked_at`	TIMESTAMPTZ	No	None	Marked when revoked.
`replay_detected_at`	TIMESTAMPTZ	No	None	Set when replay is detected.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.

Constraints: primary key on `id`; unique on `token_id`; foreign key `user_id -> users.id`. **Indexes:** `refresh_tokens_user_idx`, `refresh_tokens_family_idx`.

6. Family and Patient Master Data

6.1 families

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`family_code`	VARCHAR(30)	Yes	None	Unique family code.
`family_name`	VARCHAR(120)	Yes	None	Required family label.
`assigned`	BOOLEAN	Yes	`FALSE`	Operational assignment flag.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Constraints: primary key on `id` ; unique on `family_code` . **Index:** `families_org_assigned_idx` .

6.2 patients

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`uuid`	UUID	Yes	`gen_random_uuid()`	Unique patient UUID.
`nic`	VARCHAR(20)	No	None	Optional identity number; unique per organization when present.
`first_name`	VARCHAR(80)	Yes	None	Required.
`last_name`	VARCHAR(80)	Yes	None	Required.
`full_name`	VARCHAR(170)	Yes	Generated stored	`first_name || ' ' || last_name` .
`dob`	DATE	Yes	None	Required date of birth; set `NOT NULL` in `V7` .
`age`	SMALLINT	Yes	None	Required explicit age snapshot; set `NOT NULL` in `V7` .
`gender`	gender	Yes	None	Enum constrained.
`phone`	VARCHAR(30)	No	None	Optional phone.
`address`	TEXT	No	None	Optional address.
`blood_group`	VARCHAR(5)	No	None	Optional short blood group.
`family_id`	BIGINT	No	None	Optional foreign key to `families.id` .
`is_active`	BOOLEAN	Yes	`TRUE`	Operational active flag.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Constraints: primary key on `id` ; unique on `uuid` ; unique `patients_org_nic_unique` on `(organization_id, nic)` ; foreign key `family_id -> families.id` ; `full_name` , `dob` , and `age` are `NOT NULL` after `V7` . **Indexes:** `patients_org_full_name_idx` , `patients_org_family_idx` .

6.3 family_members

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`family_id`	BIGINT	Yes	None	Foreign key to `families.id`.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id`.
`relationship`	VARCHAR(40)	No	None	Optional relationship label.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.

Constraints: primary key on `id`; unique on `(family_id, patient_id)`; foreign keys to `families.id` and `patients.id`. **Index:** `family_members_org_family_idx`.

7. Patient Clinical Detail Tables

7.1 patient_allergies

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id`.
`allergy_name`	VARCHAR(120)	Yes	None	Required allergy label.
`severity`	allergy_severity	No	None	Enum constrained if present.
`is_active`	BOOLEAN	Yes	`TRUE`	Active flag.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Index: `patient_allergies_patient_idx`.

7.2 patient_conditions

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id`.
`condition_name`	VARCHAR(180)	Yes	None	Required condition label.
`icd10_code`	VARCHAR(16)	No	None	Optional mapped code.
`status`	VARCHAR(20)	Yes	`'active'`	String status, not enum constrained.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Index: `patient_conditions_patient_idx`.

7.3 patient_vitals

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id`.
`encounter_id`	BIGINT	No	None	Optional foreign key to `encounters.id`.
`bp_systolic`	SMALLINT	No	None	Optional systolic blood pressure.
`bp_diastolic`	SMALLINT	No	None	Optional diastolic blood pressure.
`heart_rate`	SMALLINT	No	None	Optional heart rate.

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`temperature_c`	NUMERIC(4,1)	No	None	Optional Celsius temperature.
`spo2`	SMALLINT	No	None	Optional oxygen saturation.
`recorded_at`	TIMESTAMPTZ	Yes	None	Measurement timestamp.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Index: `patient_vitals_patient_recorded_idx` on `(patient_id, recorded_at)`.

7.4 patient_timeline_events

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id`.
`encounter_id`	BIGINT	No	None	Optional foreign key to `encounters.id`.
`event_date`	DATE	Yes	None	Timeline ordering date.
`title`	VARCHAR(160)	Yes	None	Required timeline title.
`description`	TEXT	No	None	Optional narrative.
`event_kind`	VARCHAR(30)	No	None	Optional category string.
`tags`	TEXT[]	No	None	Optional tag array.
`value`	VARCHAR(80)	No	None	Optional short value.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Index: `patient_timeline_events_patient_date_idx` on `(patient_id, event_date)`.

7.5 patient_history_entries

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key; introduced in V5.
`organization_id`	UUID	Yes	None	Tenant scope.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id`.
`created_by_user_id`	BIGINT	Yes	None	Foreign key to `users.id`.
`note`	TEXT	Yes	None	Check constraint requires length between 1 and 1000.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Constraints: `patient_history_entries_note_length_chk` enforces `char_length(note) BETWEEN 1 AND 1000`. **Indexes:** `patient_history_entries_org_patient_idx`, `patient_history_entries_patient_created_idx`.

8. Scheduling and Encounter Tables

8.1 appointments

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGINT	Yes	`nextval('appointments_partitioned_id_seq')`	Sequence-backed identifier.
`organization_id`	UUID	Yes	None	Tenant scope.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id` .
`doctor_id`	BIGINT	No	None	Optional foreign key to `users.id` .
`assistant_id`	BIGINT	No	None	Optional foreign key to `users.id` .
`scheduled_at`	TIMESTAMPTZ	Yes	None	Partition key and part of primary key.
`status`	appointment_status	Yes	None	Enum constrained.
`reason`	TEXT	No	None	Optional reason text.
`priority`	priority_level	Yes	`'normal'`	Enum constrained.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Constraints: partitioned table; primary key is composite `(id, scheduled_at)` ; foreign keys to `patients.id` and optionally `users.id` for doctor/assistant. **Indexes:** `appointments_id_idx` , `appointments_status_scheduled_idx` , `appointments_patient_idx` , `appointments_org_scheduled_idx` .

Partitioning notes: V4 creates `appointments_default` and monthly partitions such as `appointments_2026_03` .

8.2 encounters

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`appointment_id`	BIGINT	Yes	None	Unique for one encounter per appointment.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id` .
`doctor_id`	BIGINT	Yes	None	Foreign key to `users.id` .
`checked_at`	TIMESTAMPTZ	Yes	None	Encounter consult time.
`notes`	TEXT	No	None	Optional notes.
`next_visit_date`	DATE	No	None	Optional follow-up date.
`status`	VARCHAR(20)	Yes	`'completed'`	String status, not enum constrained.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.
`appointment_scheduled_at`	TIMESTAMPTZ	Yes	None	Added in V4 for composite FK to appointments.

Constraints: primary key on `id` ; unique on `appointment_id` ; foreign keys to `patients.id` , `users.id` , and composite `(appointment_id, appointment_scheduled_at) -> appointments(id, scheduled_at)` with `ON UPDATE CASCADE` . **Indexes:** `encounters_patient_idx` , `encounters_appointment_fk_idx` .

8.3 encounter_diagnoses

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`encounter_id`	BIGINT	Yes	None	Foreign key to `encounters.id`.
`icd10_code`	VARCHAR(16)	No	None	Optional ICD-10 code.
`diagnosis_name`	VARCHAR(255)	Yes	None	Required diagnosis label.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.

8.4 test_orders

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`encounter_id`	BIGINT	Yes	None	Foreign key to `encounters.id`.
`test_name`	VARCHAR(180)	Yes	None	Required test label.
`status`	test_order_status	Yes	`'ordered'`	Enum constrained.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.

Index: `test_orders_encounter_idx`.

9. Prescription and Dispensing Tables

9.1 prescriptions

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`encounter_id`	BIGINT	Yes	None	Foreign key to `encounters.id`.
`patient_id`	BIGINT	Yes	None	Foreign key to `patients.id`.
`doctor_id`	BIGINT	Yes	None	Foreign key to `users.id`.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Indexes: `prescriptions_patient_idx`, `prescriptions_encounter_idx`.

9.2 prescription_items

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`prescription_id`	BIGINT	Yes	None	Foreign key to `prescriptions.id`.
`drug_name`	VARCHAR(180)	Yes	None	Required medicine label.
`dose`	VARCHAR(80)	Yes	None	Required dosing string.
`frequency`	VARCHAR(80)	Yes	None	Required frequency string.
`duration`	VARCHAR(80)	No	None	Optional duration.
`quantity`	NUMERIC(12,2)	Yes	None	Required numeric quantity.
`source`	drug_source	Yes	None	Enum constrained.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Index: `prescription_items_prescription_idx`.

9.3 dispense_records

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`prescription_id`	BIGINT	Yes	None	Foreign key to `prescriptions.id`.
`assistant_id`	BIGINT	Yes	None	Foreign key to `users.id`.
`dispensed_at`	TIMESTAMPTZ	Yes	None	Dispense timestamp.

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`status`	VARCHAR(20)	Yes	`'completed'`	String status, not enum constrained.
`notes`	TEXT	No	None	Optional notes.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.

Index: `dispense_records_prescription_idx`.

10. Inventory Tables

10.1 inventory_items

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`sku`	VARCHAR(80)	No	None	Globally unique when present.
`name`	VARCHAR(180)	Yes	None	Required item name.
`category`	inventory_category	Yes	None	Enum constrained.
`unit`	VARCHAR(20)	Yes	None	Required stock unit.
`stock`	NUMERIC(12,2)	Yes	`0`	Current stock balance.
`reorder_level`	NUMERIC(12,2)	Yes	`0`	Reorder threshold.
`is_active`	BOOLEAN	Yes	`TRUE`	Active flag.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Creation timestamp.
`updated_at`	TIMESTAMPTZ	Yes	`NOW()`	Update timestamp.
`deleted_at`	TIMESTAMPTZ	No	None	Soft delete marker.

Constraints: primary key on `id` ; unique on `sku` . **Index:** `inventory_items_org_name_idx` .

10.2 inventory_movements

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Primary key.
`organization_id`	UUID	Yes	None	Tenant scope.
`inventory_item_id`	BIGINT	Yes	None	Foreign key to `inventory_items.id` .
`movement_type`	inventory_movement_type	Yes	None	Enum constrained.
`quantity`	NUMERIC(12,2)	Yes	None	Movement quantity.
`reference_type`	VARCHAR(30)	No	None	Optional origin marker.
`reference_id`	BIGINT	No	None	Optional referenced row id.
`created_by_id`	BIGINT	Yes	None	Foreign key to `users.id` .
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Immutable creation timestamp.

Index: `inventory_movements_item_idx` .

11. Audit and Operational Tables

11.1 audit_logs

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`id`	BIGSERIAL	Yes	Sequence generated	Part of composite primary key.
`organization_id`	UUID	Yes	None	Tenant scope.

COLUMN	TYPE	REQUIRED	DEFAULT / GENERATED	CONSTRAINTS AND NOTES
`actor_user_id`	BIGINT	No	None	Optional foreign key to `users.id`.
`entity_type`	VARCHAR(60)	Yes	None	Target entity category.
`entity_id`	BIGINT	No	None	Optional target row id.
`action`	VARCHAR(30)	Yes	None	Action verb.
`ip`	INET	No	None	Optional client IP.
`user_agent`	TEXT	No	None	Optional client string.
`request_id`	UUID	No	None	Optional correlation id.
`payload`	JSONB	No	None	Optional structured payload.
`created_at`	TIMESTAMPTZ	Yes	`NOW()`	Partition key and part of composite primary key.

Constraints: partitioned by range on `created_at`; composite primary key `(id, created_at)`; foreign key `actor_user_id -> users.id`. **Indexes:** `audit_logs_created_idx`, `audit_logs_entity_idx`. **Partitions:** `audit_logs_default`, `audit_logs_2026_03`, `audit_logs_2026_04`.

12. Constraint and Validation Highlights

- **NOT NULL enforcement:** identity, relationship, and workflow-critical fields are mandatory in the schema, including `patients.first_name`, `patients.last_name`, `patients.full_name`, `patients.dob`, and `patients.age`.
- **Unique keys:** used for deduplication and natural identifiers such as user email per organization, patient NIC per organization, token id, family code, and SKU.
- **Foreign keys:** enforce patient, appointment, encounter, prescription, inventory, and actor relationships.
- **Enum validation:** several status and category fields are strictly database-constrained enums.
- **Check constraints:** explicit length validation exists on `patient_history_entries.note`.
- **Generated columns:** `patients.full_name` is database-generated and stored.
- **Partitioning:** `appointments` and `audit_logs` are partitioned for scale and maintenance.

13. Source Files

- `infra/flyway/sql/V1__init.sql`
- `infra/flyway/sql/V4__partition_appointments.sql`
- `infra/flyway/sql/V5__add_patient_history.sql`
- `infra/flyway/sql/V6__phase4_observability_security.sql`
- `infra/flyway/sql/V7__enforce_patient_required_fields.sql`
- `infra/flyway/sql/V8__seed_additional_patients.sql`
- `infra/flyway/sql/V9__add_user_extra_permissions.sql`

This dictionary is intended for review, onboarding, QA, and client walkthroughs. When the schema changes, update this document alongside the relevant Flyway migration.